

Mohamed Samir

AI & Data Science Engineer

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moknight93.github.io/Portfolio

PROFILE

Computer Science graduate with a strong foundation in software engineering and full-stack development (MERN stack), now specializing in Artificial Intelligence and Data Science. Hands-on experience across Machine Learning, Deep Learning, Computer Vision, and NLP, backed by a portfolio of applied projects. Actively focused on Data Analysis and Data Engineering to complement a growing AI/ML skill set. Seeking a junior role in AI & Data Science to apply strong CS fundamentals to real-world, data-driven problem solving.

EDUCATION

Suez Canal University

B.Sc. in Computer Science

GPA: 3.0/4.0 (Very Good)

09/2020 – 06/2024

Ismailia, Egypt

PROFESSIONAL DEVELOPMENT

Microsoft Machine Learning Engineer Trainee

Digital Egypt Pioneers Initiative - DEPI

11/2025 – 05/2026

Port Said, Egypt

- Architected end-to-end AI solutions, translating raw data pipelines into scalable data science models
- Engineered Machine Learning and Deep Learning models from scratch, applying advanced algorithms for pattern recognition across Computer Vision and Natural Language Processing (NLP).
- Implemented MLOps workflows using MLflow and Hugging Face to automate model tracking, versioning, and deployment for production-ready AI systems.
- Developed Large Language Model (LLM) solutions utilizing Prompt Engineering and Attention-based models to solve complex linguistic tasks.
- Mastered Data Preprocessing & Visualization in Python, ensuring high-quality data integrity for training and fine-tuning transfer learning models.

Deep Learning for computer vision Trainee

National Telecommunication Institute - NTI

01/2026 – 02/2026

Port Said, Egypt

- Developed and implemented Deep Learning models using Neural Networks (ANN) and Convolutional Neural Networks (CNN) for complex pattern recognition.
- Mastered Image Processing and Feature Extraction techniques to automate visual data analysis and enhance model accuracy.

- Applied Advanced Architectures including ResNet and DenseNet, utilizing Transfer Learning to optimize deep learning workflows for specialized tasks.
- Built end-to-end Computer Vision systems for Image Classification, Object Detection, and Face Recognition/Verification.
- Delivered practical, hands-on projects focused on designing and deploying object recognition systems based on modern deep learning techniques.

Data Analysis Trainee

11/2025 – 12/2025

National Telecommunication Institute - NTI

Port Said, Egypt

- Gained hands-on experience in data analysis and visualization using Excel (including Power Query), SQL Server, Power BI, and Tableau.
- Skilled in Python programming for data manipulation and analysis using NumPy, Pandas, Matplotlib, and Seaborn.
- Experienced in data preprocessing, handling, and cleaning to ensure high-quality datasets.
- Derived actionable insights and recommendations to support business decisions.
- Familiar with data modeling, ERD design, and mapping for structured data management.
- Applied data warehousing concepts to efficiently organize and retrieve large datasets.
- Focused on transforming raw data into meaningful visualizations for effective communication of results.

AI & Machine Learning Trainee

08/2025 – 09/2025

NTI-Huawei Egyptian Talents Academy (ETA)

Port Said, Egypt

- Completed the HCIA-AI V4.0 program offered by Huawei ICT Academy-Egypt in collaboration with the National Telecommunication Institute (NTI).
- Received in-depth training in Artificial Intelligence fundamentals, including machine learning and deep learning.
- Gained practical experience with Python programming, data preprocessing, TensorFlow, and model evaluation.
- Developed a strong understanding of applying AI technologies to solve real-world problems across various domains.

MERN Stack Developer Trainee

10/2024 – 04/2025

Digital Egypt Pioneers Initiative - DEPI

Port Said, Egypt

- Studied and applied MERN stack development with a focus on React front-end development.
- Gained hands-on experience with HTML, CSS, Bootstrap, Tailwind CSS, JavaScript, TypeScript, and React.
- Explored back-end technologies including Node.js, Express, NoSQL (MongoDB), and RESTful API development.
- Implemented Authentication & Authorization and performed API testing using Postman.
- Used Git/GitHub for version control and collaboration.
- Gained experience in cloud hosting and deployment using Vercel.

SKILLS

Programming Languages

- C
- C++
- Python

Back-End Technology

- Node.js
- Express.js
- MongoDB
- RESTful APIs
- Authentication & Authorization
- Postman
- FastAPI
- Flask
- Django

AI & Machine Learning

- Classical Machine Learning
- Deep Learning
- Neural Network Architectures (CNN, ResNet, DenseNet)
- Natural Language Processing (NLP)
- Large Language Models (LLMs)
- Transfer Learning
- Prompt Engineering
- PyTorch
- TensorFlow
- Keras
- Scikit-learn
- Hugging Face
- MLflow

Deployment

- Vercel
- Netlify
- Streamlit

Front-End Technology

- HTML
- CSS
- SCSS
- Bootstrap
- Tailwind CSS
- JavaScript
- TypeScript
- React.js

Data Analysis & Visualization

- Excel (Power Query)
- SQL Server
- Power BI
- Tableau
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Dash
- Plotly
- Data Cleaning & Preprocessing
- Data Modeling
- ERD Design
- Data Warehousing

Version Control

- Git
- GitHub

Prototyping

- Figma
- Framer

PROJECTS

Amazon Electronics Recommendation System

01/2026 – 05/2026

AI & ML Project

- Collaborated with a technical team to author system documentation and specifications for a 9-layer hybrid recommendation engine processing 490,000+ Amazon electronics products, covering architecture, data flow, and deployment logic.
- Documented the integration of graph-based LightGCN embeddings, sequence-aware SASRec Transformers, and ALS/BPR Matrix Factorization into a unified inference pipeline, including sub-millisecond similarity lookups via FAISS HNSW indices and LightGBM LambdaRank re-ranking.

- Translated technical implementation details into deployment guidelines and pipeline verification structures for Flask and FastAPI backend environments, and contributed scalability risk assessments addressing popularity bias via Inverse Propensity Score (IPS) debiasing.

Chest X-Ray (Pneumonia) | Image Classification & Captioning

01/2026 – 02/2026

Computer Vision Project

- Built a dual-model deep learning pipeline to ingest chest X-ray images, classify Pneumonia presence, and generate descriptive clinical text summaries.
- Designed a hybrid CNN-Encoder and LSTM-Decoder architecture, using CNN for spatial feature extraction and LSTM for sequential diagnostic text generation, with targeted data augmentation (rotation, zoom, flips) to handle dataset imbalance.
- Engineered and deployed two interfaces using Streamlit (rapid prototyping) and Flask (backend API), evaluating results via precision, recall, and F1-score for reliable, transparent clinical decision support.

Telecom Customer Churn Analysis

11/2025 – 12/2025

Data Analysis Project

- Built a relational data processing pipeline ingesting 7,043 customer transaction logs, executing custom ETL and cleaning routines in SQL and Excel to isolate high-risk behavioral indicators across demographics, subscriptions, and billing.
- Designed an interactive dashboard architecture in Power BI and Tableau, translating churn paths into visual reporting structures to safeguard a \$5.47M annual revenue pipeline.
- Analyzed multi-tiered service drop patterns, identifying month-to-month contracts and lack of technical add-ons as key attrition drivers, with 69.4% of churned accounts originating from Fiber Optic plans.

Exoplanet Hunting (NASA Space Apps Challenge Hackathon)

09/2025 – 10/2025

AI & ML Project

- Engineered an AI/ML pipeline to ingest raw, noisy time-series stellar light curves from NASA Kepler datasets to automate deep-space exoplanet detection.
- Designed a custom Convolutional Autoencoder paired with a hybrid CNN-LSTM classifier to extract spatial and temporal transit patterns while stripping cosmic noise and telemetry artifacts.
- Integrated deep-learning feature extractors with XGBoost and LightGBM in a multi-model ensemble, handling class imbalance via calculated class weights and validating robustness with precision-recall, AP, and ROC-AUC metrics.

Burns Degree Classification

08/2025 – 09/2025

AI & ML Project

- Developed a deep learning medical imaging application using TensorFlow/Keras to classify burn injury photos into 1st, 2nd, and 3rd-degree burns, processing 6,000+ medical samples with real-time augmentation and class weighting for imbalance.
- Engineered a custom CNN from scratch (77% baseline accuracy), then optimized to 79% via transfer learning with pre-trained ResNet50 and specialized classification dense blocks.

- Fine-tuned hyperparameters using Keras callbacks (EarlyStopping, ReduceLROnPlateau) and evaluated performance using medical-grade safety metrics including Precision and Recall.

Polaris | Travel Social Platform

12/2024 – 04/2025

MERN Stack Project

- Developed travel blog and social platform for users to share, discover, and interact with travel experiences worldwide.
- Built using the MERN stack (MongoDB, Express.js, React.js, Node.js) with Tailwind CSS and Bootstrap for responsive UI design.
- Implemented RESTful APIs, authentication & authorization, and cloud deployment using Vercel and GitHub.
- Designed a user-friendly interface to enhance engagement and accessibility.

Nearest Shop

12/2023 – 06/2024

Graduation Project

- Developed a location-based web and mobile ecosystem providing real-time, personalized store recommendations, bridging the gap between offline retail and digital shoppers by matching user fashion preferences with local clothing store inventories.
- Built a multi-client framework comprising a cross-platform Flutter mobile app for consumers and a responsive ReactJS web platform for administrators and shop owners to manage catalogs and monitor business metrics.
- Integrated a preference-filtering conversational chatbot and Google Cloud Vision OCR to automate user data entry and deliver personalized product recommendations based on physical attributes and category preferences.
- Orchestrated the end-to-end SDLC using Agile scrum, coordinating sprint velocities and codebase integration across an 8-engineer team, and engineered the backend (Node.js, Express, MongoDB) with OAuth 2.0 authentication and AES-secured HTTPS communication to deliver a production-ready MVP.

CERTIFICATES

- AI & Data Science Microsoft Machine Learning Engineer Certificate – Digital Egypt Pioneers Initiative (DEPI)
- AI and Machine Learning - Sprints x Microsoft Summer Camp Certificate
- Web Development - Sprints x Microsoft Summer Camp Certificate
- HCIA-AI V4.0 Certificate – Huawei ICT Academy, Egypt
- Artificial Intelligence (AI) Certificate – NTI/Huawei Egyptian Talent Academy
- React Frontend Web Developer Certificate – Digital Egypt Pioneers Initiative (DEPI)
- CS50 Introduction to Computer Science Certificate – Harvard University

LANGUAGES

Arabic

Native

English

Professional Working Proficiency (B2)

VOLUNTEERING

Video Editor

12/2023 – 06/2024

Trosc

Ismailia, Egypt

- Volunteered to create educational content in programming for university students.
- Produced and edited videos on Data Structures and Algorithms (DSA) topics, including Heap Sort, to simplify complex concepts and enhance student learning.
- Collaborated with the media team to ensure high-quality, engaging video materials.

Graphic Designer

10/2020 – 10/2022

Msh Hackers

Ismailia, Egypt

- Designed visual content to support educational programming initiatives within the university team.
- Created graphics that improved the presentation and clarity of learning materials.
- Contributed to the overall quality and appeal of educational resources through creative design solutions.